

PROBLEMS ON RANDOM ACCESS PROTOCOLS

Q.1. A group of N stations share 100 Kbps slotted ALOHA channel. Each station output a 500 bits frame on an average of 5000 ms even if previous one has not been sent. What is the required value of N?

Throughput of One Station-

Throughput of each station
= Number of bits sent per second
= 500 bits / 5000 m
= 500 bits / (5000 x 10^{-3} sec)
= 100 bits/sec

Throughput of Slotted Aloha-

Throughput of slotted aloha
= Efficiency x Bandwidth
= $0.368 \times 100 \text{ Kbps}$
= 36.8 Kbps

Total Number of Stations-

Throughput of slotted aloha = Total number of stations x Throughput of each station

Substituting the values, we get-

$$36.8 \text{ Kbps} = N \times 100 \text{ bits/sec}$$

$$\therefore N = 368$$

Thus, required value of $N = 368$.

Q.2. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA / CD. The signal travels along the wire at 2×10^8 m/sec. What is the minimum packet size that can be used on this network?

Given-

Distance = 2 km

Bandwidth = 10^7 bps

Speed = 2×10^8 m/sec

Calculating Propagation Delay-

Propagation delay (T_p)

= Distance / Propagation speed

= 2 km / (2×10^8 m/sec)

= 2×10^3 m / (2×10^8 m/sec)

= 10^{-5} sec

Calculating Minimum Frame Size-

Minimum frame size

= 2 x Propagation delay x Bandwidth

= $2 \times 10^{-5} \text{ sec} \times 10^7 \text{ bits per sec}$

= 200 bits or 25 bytes

PROBLEMS ON SLIDING WINDOW PROTOCOLS

1. If the bandwidth of the line is 1.5 Mbps, RTT is 45 msec and packet size is 1 KB, then find the link utilization in stop and wait.

Given-

Bandwidth = 1.5 Mbps

RTT = 45 msec

Packet size = 1 KB

Calculating Transmission Delay-

Transmission delay (T_t)

= Packet size / Bandwidth

= 1 KB / 1.5 Mbps

= $(2^{10} \times 8 \text{ bits}) / (1.5 \times 10^6 \text{ bits per sec})$

= 5.461 msec

Calculating Propagation Delay-

$$\begin{aligned} &\text{Propagation delay (T}_p\text{)} \\ &= \text{Round Trip Time} / 2 \\ &= 45 \text{ msec} / 2 = 22.5 \text{ msec} \end{aligned}$$

Calculating Value of 'a'-

$$\begin{aligned} a &= T_p / T_t \\ a &= 22.5 \text{ msec} / 5.461 \text{ msec} \\ a &= 4.12 \end{aligned}$$

Calculating Link Utilization-

$$\begin{aligned} &\text{Link Utilization or Efficiency } (\eta) \\ &= 1 / 1+2a \\ &= 1 / (1 + 2 \times 4.12) = 1 / 9.24 = 0.108 = 10.8 \% \end{aligned}$$

2. A channel has a bit rate of 4 Kbps and one way propagation delay of 20 msec. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be-

80 bytes

80 bits

160 bytes

160 bits

Given-

Bandwidth = 4 Kbps

Propagation delay (T_p) = 20 msec

Efficiency $\geq 50\%$

Let the required frame size = L bits.

Calculating Transmission Delay

Transmission delay (T_t)

= Packet size / Bandwidth

= L bits / 4 Kbps

Calculating Value Of 'a'-

$$a = T_p / T_t$$

$$a = 20 \text{ msec} / (L \text{ bits} / 4 \text{ Kbps})$$

$$a = (20 \text{ msec} \times 4 \text{ Kbps}) / L \text{ bits}$$

Condition For Efficiency To Be At least 50%-

For efficiency to be at least 50%, we must have-

$$1 / (1 + 2a) \geq 1/2$$

$$a \leq 1/2$$

Substituting the value of 'a', we get-

$$(20 \text{ msec} \times 4 \text{ Kbps}) / L \text{ bits} \leq 1/2$$

$$L \text{ bits} \geq (20 \text{ msec} \times 4 \text{ Kbps}) \times 2$$

$$L \text{ bits} \geq (20 \times 10^{-3} \text{ sec} \times 4 \times 10^3 \text{ bits per sec}) \times 2$$

$$L \text{ bits} \geq 20 \times 4 \text{ bits} \times 2$$

$$L \geq 160$$

From here, frame size must be at least 160 bits.

Thus, Correct Option is (D).

3. A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the “go back n ARQ” scheme with n set to 10. Assuming that each frame is 100 bytes long, what is the maximum data rate possible?

5 Kbps

10 Kbps

15 Kbps

20 Kbps

Given-

Bandwidth = 20 Kbps

Propagation delay (T_p) = 400 ms

Frame size = 100 bytes

Go back N is used where $N = 10$

Calculating Transmission Delay-

Transmission delay (T_t)

= Frame size / Bandwidth

= 100 bytes / 20 Kbps

= (100 x 8 bits) / (20 x 10^3 bits per sec)

= 0.04 sec

= 40 msec

Calculating Value Of 'a'-

$a = T_p / T_t$

$a = 400 \text{ msec} / 40 \text{ msec}$

$a = 10$

Calculating Efficiency-

Efficiency (η)

$$= N / (1+2a)$$

$$= 10 / (1 + 2 \times 10)$$

$$= 10 / 21$$

$$= 0.476$$

$$= 47.6 \%$$

Calculating Maximum Data Rate Possible-

Maximum data rate possible or Throughput

$$= \text{Efficiency} \times \text{Bandwidth}$$

$$= 0.476 \times 20 \text{ Kbps}$$

$$= 9.52 \text{ Kbps}$$

$$\cong 10 \text{ Kbps}$$

Thus, Correct Option is (B)

4. Consider a 128×10^3 bits/sec satellite communication link with one way propagation delay of 150 msec. Selective Retransmission (repeat) protocol is used on this link to send data with a frame size of 1 KB. The minimum number of sequence numbers to achieve 100% utilization is _____ .

Given-

Bandwidth = 128×10^3 bits/sec

Propagation delay (T_p) = 150 msec

Frame size = 1 KB

Now,

To achieve 100% utilization, efficiency must be 100%.

Efficiency is 100% when sender window size is optimal i.e. $1+2a$

Calculating Transmission Delay-

Transmission delay (T_t)

= Frame size / Bandwidth

= 1 KB / (128×10^3 bits per sec)

= $(1 \times 2^{10} \times 8 \text{ bits}) / (128 \times 10^3 \text{ bits per sec})$

= 64 msec

Calculating Value of 'a'-

$$a = T_p / T_t$$

$$a = 150 \text{ msec} / 64 \text{ msec}$$

$$a = 2.34$$

Calculating Optimal Sender Window Size-

Optimal sender window size

$$= 1 + 2a$$

$$= 1 + 2 \times 2.34$$

$$= [5.68]$$

$$= 6$$

Calculating Number Of Sequence Numbers Required-

In SR Protocol, sender window size and receiver window size are same.

So, sender window size = receiver window size = 6

For any sliding window protocol, minimum number of sequence numbers required

= Sender window size + Receiver window size

$$= 6 + 6$$

$$= 12$$

Example 24.1

The following is the contents of a UDP header in hexadecimal format.

CB84000D001C001C

- a.** What is the source port number?
- b.** What is the destination port number?
- c.** What is the total length of the user datagram?
- d.** What is the length of the data?
- e.** Is the packet directed from a client to a server or vice versa?
- f.** What is the client process?

Solution

- a) The source port number is the first four hexadecimal digits (CB84)₁₆. Which means that the source port number is 52100.
- b) The destination port number is the second four hexadecimal digits (000D)₁₆.
Which means that the destination port number is 13
- c) The third four hexadecimal digits (001C)₁₆ define the length of the whole UDP packet as 28 bytes.
- d) The length of the data is the length of whole packet minus the length of header i. e $28 - 8 = 20$ bytes
- e) Since the destination port number is 13 (well known port), the packet is from the client to the server.
- f) The client process is the Daytime.

Example 1

- Imagine a TCP connection is transferring a file of 6000 bytes. The first byte is numbered 10010. What are the sequence numbers for each segment if data are sent in five segments with the first four segments carrying 1000 bytes and the last segment carrying 2000 bytes?

- **Solution:**

The following shows the sequence number for each segment:

Segment 1 ==> sequence number: 10,010 (range: 10,010 to 11,009)

Segment 2 ==> sequence number: 11,010 (range: 11,010 to 12,009)

Segment 3 ==> sequence number: 12,010 (range: 12,010 to 13,009)

Segment 4 ==> sequence number: 13,010 (range: 13,010 to 14,009)

Segment 5 ==> sequence number: 14,010 (range: 14,010 to 16,009)

Example 23.2.4

The following is a dump of a TCP header in hexadecimal format

E2930017 00000001 00000000 500207FF 00000000 ...

What is the source port number?

What is the destination port number?

What is sequence number?

What is the acknowledgment number?

What is the length of the header?

What is the type of the segment?

What is the window size?

Answer

- a. The source port number is **(E293)₁₆** or **58,003** in decimal.
- b. The destination port number is **(0017)₁₆** or **23** in decimal.
- c. The sequence number is **(00000001)₁₆** or **1** in decimal.
- d. The acknowledgment number is **(00000000)₁₆** or **0** in decimal.
- e. The HLEN = **5**. The header is **5x4 = 20 bytes** long.
- f. The combination of the reserved field and the control field is **(002)₁₆** or **(000000000010)₂**. The rightmost 6 bits are **000010**, which means only the SYN bit is set. This is the SYN segment used for connection establishment.
- g. The window size field is **(07FF)₁₆** or **2047** in decimal. The window size is **2047 bytes**.

Consider the effect of using slow start on a line with a 10 msec RTT and no congestion. The receiver window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?

Given

- Receiver window size = 24 KB
- Maximum Segment Size = 2 KB
- RTT = 10 msec

Sol

- Receiver window size in terms of MSS
- = Receiver window size / Size of 1 MSS
- = 24 KB / 2 KB
- = 12 MSS

- Slow Start Threshold-
- = Receiver window size / 2
- = 12 MSS / 2
- = 6 MSS

Slow Start Phase-

- Window size at the start of 1st transmission = 1 MSS
- Window size at the start of 2nd transmission = 2 MSS
- Window size at the start of 3rd transmission = 4 MSS
- Window size at the start of 4th transmission = 6 MSS

Since the threshold is reached, so it marks the end of slow start phase.

Now, congestion avoidance phase begins.

Congestion Avoidance Phase-

- Window size at the start of 5th transmission = 7 MSS
- Window size at the start of 6th transmission = 8 MSS
- Window size at the start of 7th transmission = 9 MSS
- Window size at the start of 8th transmission = 10 MSS
- Window size at the start of 9th transmission = 11 MSS
- Window size at the start of 10th transmission = 12 MSS

From here,

- Window size at the end of 9th transmission or at the start of 10th transmission is 12 MSS.
- Thus, 9 RTT's will be taken before the first full window can be sent.

So,

Time taken before the first full window is sent

= 9 RTT's

= 9 x 10 msec

= 90 msec

Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of slow start phase is 2 MSS and the threshold at the start of first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of tenth transmission.

1. 8 MSS
2. 14 MSS
3. 7 MSS
4. 12 MSS

Solution-

Given-

- Window size at the start of slow start phase = 2 MSS
- Threshold at the start of first transmission = 8 MSS
- Time out occurs during 5th transmission

Slow Start Phase-

- Window size at the start of 1st transmission = 2 MSS
- Window size at the start of 2nd transmission = 4 MSS
- Window size at the start of 3rd transmission = 8 MSS

Since the threshold is reached, so it marks the end of slow start phase.

Now, congestion avoidance phase begins.

Congestion Avoidance Phase-

- Window size at the start of 4th transmission = 9 MSS
- Window size at the start of 5th transmission = 10 MSS

It is given that time out occurs during 5th transmission.

TCP reacts by-

- Setting the slow start threshold to half of the current congestion window size.
- Decreasing the congestion window size to 2 MSS (Given value is used).
- Resuming the slow start phase.

So now,

- Slow start threshold = $10 \text{ MSS} / 2 = 5 \text{ MSS}$
- Congestion window size = 2 MSS

Slow Start Phase-

- Window size at the start of 6th transmission = 2 MSS
- Window size at the start of 7th transmission = 4 MSS
- Window size at the start of 8th transmission = 5 MSS

Since the threshold is reached, so it marks the end of slow start phase.

Now, congestion avoidance phase begins.

Congestion Avoidance Phase-

- Window size at the start of 9th transmission = 6 MSS
- Window size at the start of 10th transmission = 7 MSS
- Window size at the start of 11th transmission = 8 MSS

From here,

Window size at the end of 10th transmission

= Window size at the start of 11th transmission

= 8 MSS

Thus, Option (A) is correct.

Suppose that the TCP congestion window is set to 18 KB and a time out occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the MSS is 1 KB.

Solution-

Congestion Window Size-

Congestion window size in terms of MSS

= 18 KB / Size of 1 MSS

= 18 KB / 1 KB

= 18 MSS

Reaction Of TCP On Time Out-

TCP reacts by-

- Setting the slow start threshold to half of the current congestion window size.
- Decreasing the congestion window size to 1 MSS.
- Resuming the slow start phase.

So now,

- Slow start threshold = $18 \text{ MSS} / 2 = 9 \text{ MSS}$
- Congestion window size = 1 MSS

Slow Start Phase-

- Window size at the start of 1st transmission = 1 MSS
- Window size at the start of 2nd transmission = 2 MSS
- Window size at the start of 3rd transmission = 4 MSS
- Window size at the start of 4th transmission = 8 MSS
- Window size at the start of 5th transmission = 9 MSS

Thus, after 4 successful transmissions, window size will be 9 MSS or 9 KB.